

Final report

Viral Pneumonia and COVID-19 detection using chest x-ray images

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1. Introduction

The recent global health crisis precipitated by COVID-19 has underscored the critical need for rapid and accurate diagnostic tools. Among the various approaches explored, machine learning models applied to radiographic imaging, such as chest X-rays, have a potential in aiding the early detection of respiratory conditions, including COVID-19 and Viral Pneumonia. This final report presents the best machine learning algorithms to determine the most effective model for detecting these diseases using lung X-ray images. The ultimate objective is to develop an algorithm for deployment in clinical settings, offering doctors an additional diagnostic tool that enhances their decision-making process.

The data used in this analysis is sourced from the publicly available COVID-19 Radiography Database on Kaggle. This dataset comprises thousands of chest X-ray images classified into four categories: Normal, COVID-19, Lung Opacity, and Viral Pneumonia. Lung Opacity is a degree of haziness referred to as opacity in the X-ray, which may indicate a potential disease other than COVID-19 or Viral Pneumonia. The dataset's diversity in image quality and pathological variations makes it well-suited for developing robust diagnostic algorithms.

2. Aim

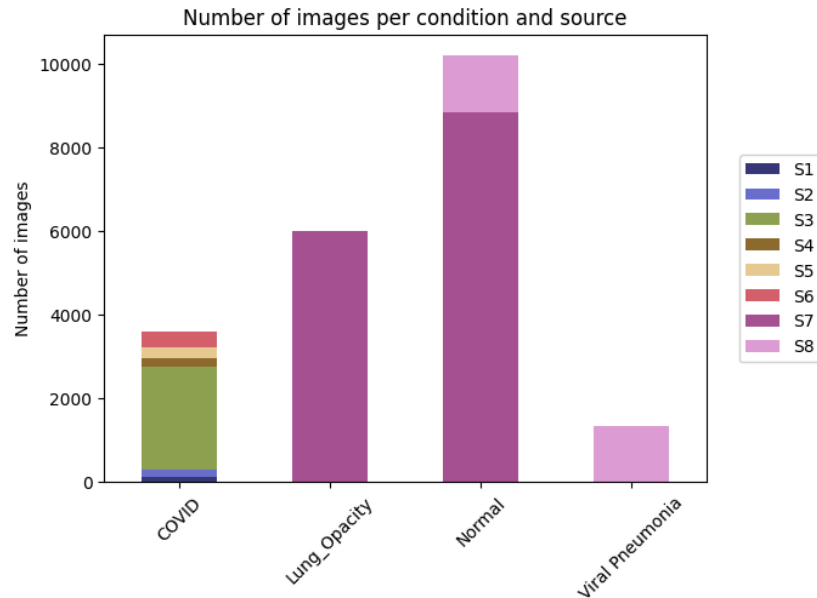
The primary objective is to identify the most effective machine learning model for detecting COVID-19 and Viral Pneumonia from chest X-rays. The selected model should excel at identifying these conditions, ensuring high recall rates to minimize the risk of false negatives. This is especially critical in a healthcare context, where the consequences of missing a diagnosis can be severe.

Accordingly, the evaluation will focus on maximizing the F1 Score for the COVID-19 and Viral Pneumonia classes, with a particular emphasis on recall as the key component. While precision is also important, the priority is to avoid false negatives—instances where a patient is wrongly classified as healthy. In addition to these metrics, we also consider computational efficiency, the minimization of overfitting, and the interpretability of the model to ensure it is accurately detecting the lungs as the relevant area in the images.

3. Exploratory data analysis

The dataset was obtained from the COVID-19 Radiography Database that is available on Kaggle ([COVID-19 Radiography Database](#)). It consists of chest X-ray (CXR) images (299x299) and corresponding lung masks (256x256), both in grayscale, collected from various public sources. The images are of high quality, as all files are readable and correctly labeled, with no corrupted or mis-sized images and only a small number of duplications. A total of 21,165 images are categorized into four classes: Normal (10,192), COVID-19 (3,616), Lung Opacity (6,012), and Viral Pneumonia (1,345) (Figure 3.1). Additionally, we observed that lung opacity and viral pneumonia images are sourced from a single location each, while normal and COVID-19 images are sourced from multiple locations. The different sources are specified in Table 3.1. Many of them are collections of images themselves.

Figure 3.1: Composition of the dataset



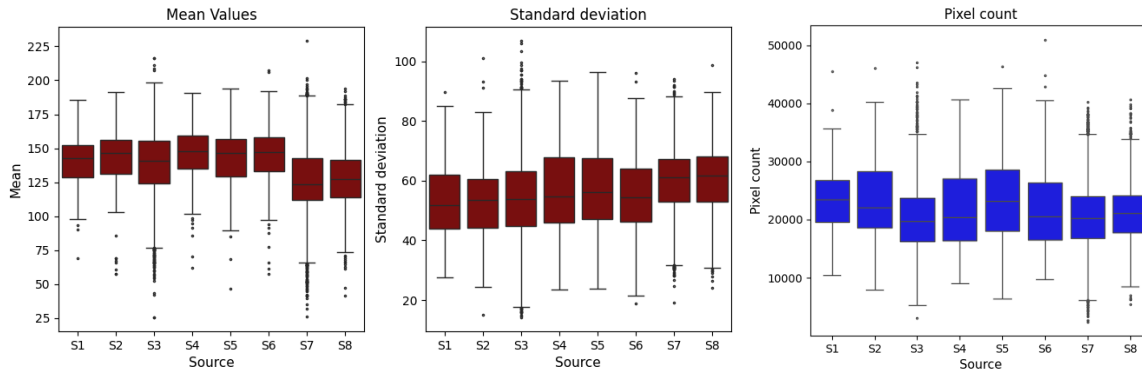
Next, we examined whether X-ray images varied by source, focusing on systematic differences in image coloring and lung size. We assessed several statistical variables, including pixel count, mean, standard deviation, and other histogram metrics.

Our findings revealed significant differences in mean pixel intensity, standard deviation, and lung size among images from different sources (Figure 3.2). ANOVA tests confirmed these differences, indicating that the images from some sources have notably different mean pixel values, standard deviations, and pixel counts. The differences of the images across the sources should help in the training of the model to improve its ability to generalize.

Table 3.1: Overview of the image sources

Source	URL
S1 Italian Society of Radiology	https://sirm.org/category/senza-categoria/covid-19/
S2 COVID-19 image data collection	https://github.com/ieee8023/covid-chestxray-dataset
S3 BIMCV-COVID	https://bimcv.cipf.es/bimcv-projects/bimcv-covid19/#1590858128006-9e640421-6711
S4 Hannover Medical School	https://github.com/ml-workgroup/covid-19-image-repository/tree/master/png
S5 Eurorad	https://eurorad.org
S6 COVID-CXNet	https://github.com/armiro/COVID-CXNet
S7 RSNA Pneumonia Detection Challenge	https://www.kaggle.com/c/rsna-pneumonia-detection-challenge/data
S8 Guangzhou Women and Children's Medical Center	https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia

Figure 3.2: Boxplots illustrating the distribution of mean pixel intensity, standard deviation, and lung size (pixel count of the lung area) per source

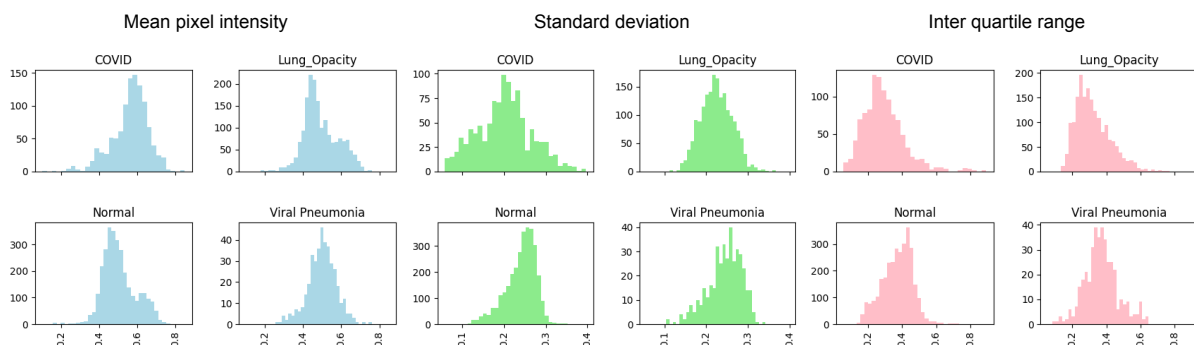


So far our analysis showed an imbalanced class distribution and a bias towards a particular source for each of the four classes. In addition, we found that the images' optical properties and lung sizes are influenced by their source.

There are several concerns that arise from these observations. Firstly, the class imbalance may lead to biased model training, where the model becomes more proficient at identifying the overrepresented classes, such as Normal, while underperforming on the less represented ones, like Viral Pneumonia. Secondly, the bias towards specific sources for each class could introduce variability in image characteristics, such as differences in image quality, brightness, or texture. This variability may affect the model's ability to generalize across different sources, potentially resulting in reduced performance when encountering images from less-represented sources. Additionally, there is a risk that the model could overfit to the dominant classes or sources, which might decrease its effectiveness in real-world scenarios. To address these concerns, it is essential to test and employ strategies such as data augmentation, resampling, and normalization/standardization, and to carefully evaluate the model to ensure its robustness and generalizability.

In addition, we analyzed the class-dependent optical properties to gain insights into how the pixel intensity distributions vary between the classes. The four classes exhibit distinct distributions of pixel intensity statistics (Figure 3.3).

Figure 3.3: Histograms showing the distribution of mean pixel intensity (blue), standard deviation of pixel intensity (green) and inter quartile range (red) per class



COVID-19 cases tend to have a higher mean pixel intensity, more spread-out standard deviation, and a lower interquartile range (IQR) compared to other classes. While the COVID-19 class shows a clear distinction, the respective distributions for normal and viral

pneumonia look more similar. The strongest similarity of optical properties is observed between the distributions of lung opacity and normal cases. This might make it more challenging for models to distinguish between these two categories. It is noteworthy that most images from these two classes originate from a common source, which raises the possibility that the observed differences between the classes might stem from variations in image properties linked to the source. These observations emphasize again the need for strategic image preprocessing including normalization or standardization and data augmentation.

4. Modelisation

For the detection of COVID-19 and Viral Pneumonia from chest X-rays, we began with classic machine learning techniques, including Support Vector Machines (SVM) and Random Forests. Building on this foundation, we advanced to deep learning methods, particularly Convolutional Neural Networks (CNNs), to enhance detection accuracy and manage the dataset's complexity. The results of both approaches will be presented in the following chapters.

4.1. Classical machine learning

In the initial phase, we used classical machine learning models to detect COVID-19 and Viral Pneumonia from chest X-rays. Several preprocessing baselines (Table 4.1) were established to assess the impact of normalization, standardization, image size, and masking on model performance.

Table 4.1: Overview of preprocessing operations applied to baselines

		Baseline					
		1.0	1.1	1.2	1.3	1.4	2.0
Applied pre-processing operation	Grayscale	X	X	X	X	X	X
	Normalized	X	X		X		
	Min-Max Normalized			X			
	Resized (64, 64)	X		X	X	X	
	Standardized					X	X
	Masked				X		X

The basic models tested included Logistic Regression, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees. Each model was configured with specific parameters: Logistic Regression used the "lbfgs" solver and C=1.0, SVM had a gamma value of 0.01 and a polynomial kernel, KNN utilized 7 neighbors with the Minkowski metric, and Decision Trees were set with the entropy criterion and a maximum depth of 4. Among these models, SVM and KNN performed the best across baselines judging by the average F1 score and the F1 score of the COVID class which was more challenging to detect.

Looking at the different preprocessing baselines, we observed the most favorable results with baseline 1.4 indicating that standardization of the data improved model performance. In addition, we learned that model performance did not benefit from masking or upsizing the figures.

Dimension reduction and resampling methods were also explored but ultimately discarded due to minimal improvement and the introduction of additional noise in the data. Instead, we refined the hyperparameters for SVM and KNN using GridSearch to optimize their performance. In addition to these models, we tested more advanced algorithms such as Random Forest and XGBoost on baseline 1.4.

The two best classical machine learning models identified were XGBoost and GridSearch-optimized SVM (gSVM). Both models demonstrated strong overall performance (summarized in Table 4.2), with XGBoost achieving a macro average F1-Score of 0.87 and gSVM slightly higher at 0.88.

Table 4.2: Performance of the optimized Gridsearch SVM (gSVM) and XGBoost model using preprocessing baseline 1.4

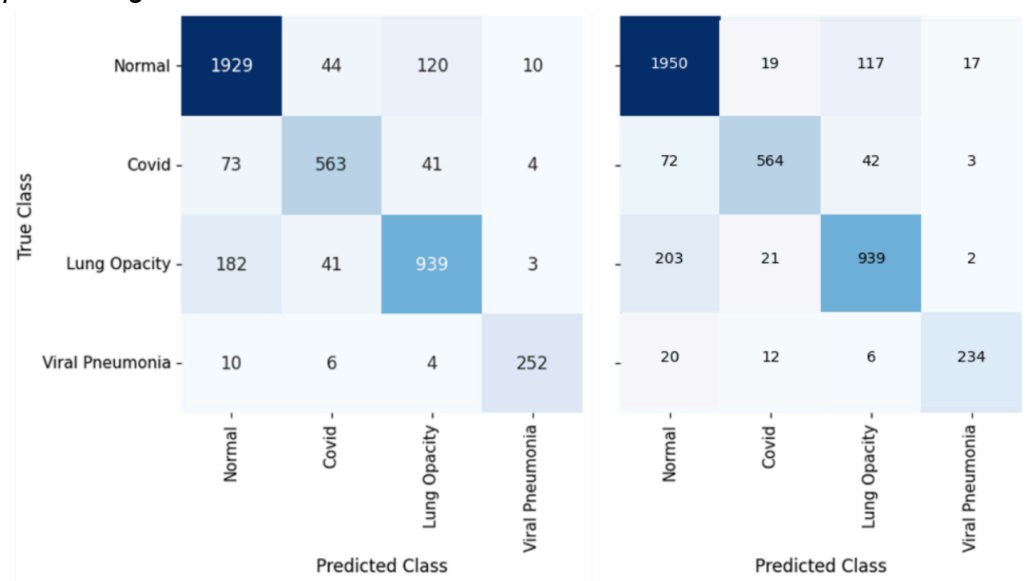
	accuracy score	F1- Score				
		Normal	COVID	Lung Opacity	Viral Pneumonia	macro average
gSVM	0.87	0.9	0.84	0.83	0.93	0.88
XGBoost	0.87	0.9	0.87	0.83	0.89	0.87

gSVM excelled in identifying the Normal and Viral Pneumonia classes, while XGBoost performed better in detecting COVID-19, with an F1-Score of 0.87 compared to gSVM's 0.84.

The recall rate for COVID-19 detection is 0.83, reflecting its effectiveness at minimizing missed COVID-19 cases, which is vital for accurate diagnosis. gSVM's recall for COVID-19 stands slightly lower at 0.83.

In Figure 4.1 we depict the confusion matrices for the gSVM model and XGBoost, showing the distribution of true positive and false positive classifications. We observe that both models classified the majority of COVID cases correctly (563 instances for gSVM and 564 XGBoost) while misclassification of lung opacity as normal and vice versa was the most common mistake for both models. This is not surprising as these two classes showed most similar optical properties during exploratory data analysis.

Figure 4.1: Confusion Matrix of GridSearch optimized SVM (gSVM, left) and XGBoost (right) for preprocessing baseline 1.4



In terms of computational efficiency, XGBoost offers an additional advantage. With an execution time of 15 minutes, it is faster than gSVM, which takes 22 minutes. This makes XGBoost a more practical option for real-time applications, especially in clinical environments where timely results are critical.

In summary, both models show strengths in different areas. XGBoost is superior in COVID-19 detection and computational efficiency, making it the preferred choice for real-time clinical applications. gSVM, while slightly slower and less effective at detecting COVID-19, excels in identifying Viral Pneumonia and Normal cases, offering strong overall performance as a viable alternative

4.2. Deep learning

Deep learning techniques are at the forefront of enhancing the accuracy and efficiency of COVID-19 detection from chest X-ray images. This section focuses on applying these advanced methods to explore various deep learning architectures, with the goal of identifying the most effective models for reliably classifying images into four categories: normal, COVID-19, lung opacity, and viral pneumonia. This involves not only evaluating the performance of different models but also understanding the specific characteristics and features that make certain models more successful than others in this task.

4.2.1. Methodology

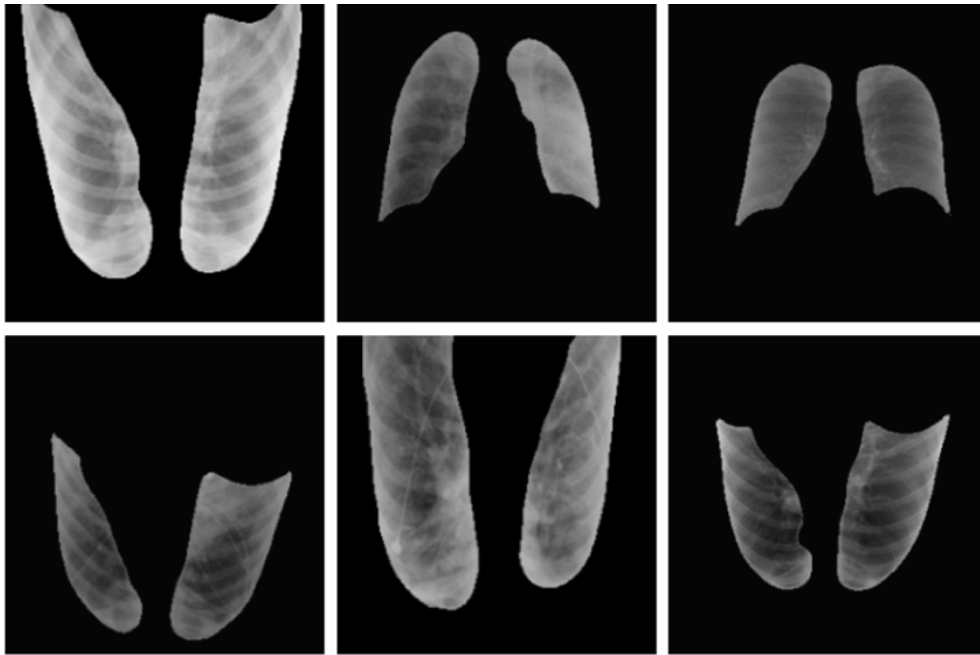
The methodology encompasses data collection and preprocessing, experimentation with diverse deep learning architectures, and comprehensive evaluation metrics to optimize and validate the classification of chest X-ray images.

4.2.1.1. Data Collection and Preprocessing

For this analysis, we utilized the publicly available dataset from Kaggle, which contains chest X-ray images categorized into four classes: normal, COVID-19, lung opacity, and viral pneumonia. After downloading the dataset, we organized the images and associated masks into a structured format, enabling efficient access and processing during model training. The dataset was then split into training and validation sets, maintaining a balanced distribution across the four categories to ensure robust evaluation.

Preprocessing steps were crucial in preparing the data for model training. We experimented with converting images to grayscale, using either three channels or just one to simplify the data and reduce computational complexity. The image sizes were typically either 64x64 or full-size, depending on the model, with transfer learning requiring specific input sizes. Data augmentation techniques, such as rotation, flipping, and scaling, were also tested to increase the diversity of the training data. For instance, our augmentation function randomly flips the weighted image both horizontally and vertically with a 50% chance, rotates it by up to 10 degrees, and shifts the image dimensions by up to 10%. We also incorporated random zooming (up to 10%) and adjusted the brightness and contrast of the images. Some models utilized these augmentation techniques, while others did not.

Figure 4.2: Example illustrations showing X-ray images after masking and data augmentation



Additionally, we applied masks available in the Kaggle dataset by weighting the images with the masks, multiplying the binary mask with the image to focus the model on specific regions of interest. This experimentation with different preprocessing combinations was essential for determining the most effective approach to enhancing model accuracy and generalization.

4.2.1.2. Model Selection Process

In our model selection process, we explored a variety of deep learning architectures to identify the most effective approach for classifying X-ray images into four categories: normal, COVID-19, lung opacity, and viral pneumonia. The models are summarized in Table 4.3.

We began by investigating a hybrid approach that combined the strengths of Convolutional Neural Networks (CNNs) for feature extraction with Support Vector Machines (SVM) for classification. This architecture utilized three layers, leveraging outputs from a pre-trained base model and incorporating a GlobalAveragePooling2D layer.

Next, we employed transfer learning techniques, fine-tuning three pre-trained models, ResNet50, EfficientNetB1, and VGG16. These models were adapted to our dataset with additional layers tailored to the specific classification task. A typical architecture comprised eight layers that included a mix of dense layers, batch normalization, dropout, and ReLU activation functions. To optimize training, we implemented callback functions such as Early Stopping and ReduceLROnPlateau.

Table 4.3: Overview of tested models

Model	Architecture	Activation Function	Batch Size	Callback Functions
Hybrid Model	3 Layers: 1 Base Model Output (from pre-trained model), 1 GlobalAveragePooling2D, 1 Feature Model Output	-	-	-
Transfer Learning	8 Layers: 1 Base Model Output (from pre-trained model), 1 GlobalAveragePooling2D, 1 Dense (1024 units, ReLU, L2), 1 BatchNorm, 1 Dropout (0.2), 1 Dense (512 units, ReLU, L2), 1 BatchNorm, 1 Dropout (0.2), 1 Dense (256 units, ReLU, L2), 1 BatchNorm, 1 Dropout (0.2), 1 Output (4 units, softmax)	ReLU	32	Early Stopping, ReduceLROnPlateau
DNN 1.0	4 Dense Layers: 1 Dense (256 units, ReLU), 1 Dense (128 units, ReLU), 1 Dense (64 units, ReLU), 1 Dense (4 units, softmax)	ReLU	200	-
DNN 1.1	7 Dense Layers: 1 Dense ((weight*height)/4 units, ReLU), 1 Dense ((weight*height)/16 units, ReLU), 1 Dense ((weight*height)/64 units, ReLU), 1 Dense (256 units, ReLU), 1 Dense (128 units, ReLU), 1 Dense (64 units, ReLU), 1 Dense (4 units, softmax)	ReLU	200	-
CNN 1.0	9 Layers: 1 Conv2D (32 filters, 5x5 kernel, ReLU), 1 MaxPooling, 1 Dropout, 1 Flatten, 1 Dense (256 units, ReLU), 1 Dense (128 units, ReLU), 1 Dense (64 units, ReLU), 1 Dense (32 units, ReLU), 1 Dense (16 units, ReLU), 1 Output (4 units, softmax)	ReLU	100	Early Stopping, ReduceLROnPlateau
CNN 1.1	15 Layers: 1 Input (256x256x1), 1 Conv2D (32 filters, 3x3 kernel, ReLU, L2), 1 BatchNorm, 1 MaxPooling, 1 Conv2D (64 filters, 3x3 kernel, ReLU, L2), 1 BatchNorm, 1 MaxPooling, 1 Conv2D (128 filters, 3x3 kernel, ReLU, L2), 1 BatchNorm, 1 MaxPooling, 1 Conv2D (256 filters, 3x3 kernel, ReLU, L2), 1 BatchNorm, 1 MaxPooling, 1 Conv2D (512 filters, 3x3 kernel, ReLU, L2), 1 BatchNorm, 1 MaxPooling, 1 Flatten, 1 Dense (512 units, ReLU, L2), 1 Dropout (0.5), 1 Dense (256 units, ReLU, L2), 1 Dropout (0.5), 1 Output (4 units, softmax)	ReLU	64	Early Stopping, ReduceLROnPlateau

In addition, we tested two different Deep Neural Network (DNN) architectures, labeled DNN 1.0 and DNN 1.1. DNN 1.0 featured a straightforward four-layer dense network, while DNN 1.1 presented a more complex structure with seven layers, characterized by progressively smaller dense layers according to the image's height and width.

We also designed and evaluated two Convolutional Neural Network (CNN) architectures, designated CNN 1.0 and CNN 1.1. CNN 1.0 was relatively simple, integrating convolutional layers, max pooling, dropout, and dense layers. In contrast, CNN 1.1 was more elaborate, comprising 15 layers and incorporating batch normalization and L2 regularization to enhance performance.

Throughout the model selection process, we experimented with various factors, such as using masked versus unmasked images, testing different image sizes, and comparing grayscale with RGB inputs. We also tested different activation functions and batch sizes to evaluate their effects on performance. Additionally, selective data augmentation was applied to assess its impact on model performance.

4.2.1.3. Experimental Setup

The experiments were conducted using a standard training and validation split of 80/20 to assess the models' ability to generalize to unseen data. The dataset comprised a total of

21,165 X-ray images, unevenly distributed across the four categories: normal, COVID-19, lung opacity, and viral pneumonia, sourced from the Kaggle dataset.

Training was carried out on Google Colab Pro Plus, utilizing its computational resources. The batch size ranged from 32 to 200, with a starting learning rate of 0.001. Most experiments employed callbacks, including ReduceLROnPlateau and Early Stopping, to optimize training and prevent overfitting. The maximum number of epochs was set to 50 for most models.

The implementation relied on several key libraries, including TensorFlow and Keras for building and training deep learning models, NumPy and Pandas for data manipulation, and Matplotlib and Seaborn for visualizing results.

4.2.1.4. Criteria for Model Selection

The experiments were conducted using a standard training and validation split to assess the models' ability to generalize to unseen data. The evaluation of the models was based on several criteria, including F1-scores for all categories, which provided a balanced view of precision and recall, and recall specifically for COVID-19 cases, highlighting the importance of accurately identifying this critical condition. Overall accuracy was also considered to reflect general performance across all categories, along with execution time to evaluate the practicality of each model for real-time or large-scale applications. Additionally, we utilized Grad-CAM to provide visual insights into the regions of the images that contributed to the model's predictions, enhancing our understanding of model behavior. These metrics enabled us to select models that not only demonstrated strong performance but were also efficient and suitable for clinical use.

4.2.2. Broad Results - Initial Model Experiments

The results of the evaluation of various model configurations for detecting COVID and other lung conditions are displayed in Table 4.4.

4.2.2.1. Model Architecture

The results from the various model architectures can be grouped into five main categories: hybrid models, transfer learning models, DNNs, CNN 1.0, and CNN 1.1. Each category demonstrates unique characteristics and performance outcomes.

The hybrid approach, combining ResNet50 with an SVM classifier, shows moderate performance, with a clear distinction between masked and unmasked data. The unmasked, no DA, RGB, 224x224 configuration outperforms its masked counterpart, achieving an accuracy of 0.91 compared to 0.87, and higher F1-scores across all classes. This suggests that the hybrid model benefits from retaining full image information without masking.

Transfer learning models generally perform very well, particularly when data augmentation is applied. For example, the unmasked, DA, RGB, 240x240 configuration of EfficientNetB1 achieved the highest accuracy of 0.94, demonstrating the strength of using unmasked images in these architectures. The masked versions, while still performing robustly, did not reach the same level of accuracy, with EfficientNetB1 scoring 0.90. This indicates that, similar to the hybrid models, unmasked data provides a slight edge in performance for transfer learning models. Overall, both VGG16 and ResNet50, alongside EfficientNetB1, show that transfer learning is a powerful approach, particularly when combined with data augmentation, leading to strong performance in accuracy and F1-scores across different configurations.

The DNN models display the weakest performance among the groups, particularly when working with larger image sizes. The unmasked, no DA, Gray, 64x64 configuration of DNN 1.0 achieved an accuracy of 0.83, which decreased slightly with the larger 299x299 image size, suggesting potential overfitting or that the architecture is not well-suited to leverage larger images. The extended version, DNN 1.1, shows marginal improvement but still underperforms compared to CNN-based architectures, particularly in computational efficiency and overall accuracy. This underperformance highlights that DNNs, in general, are not well-suited for image classification tasks, especially when dealing with complex medical images like X-rays. The architecture lacks the capacity to effectively capture spatial hierarchies and local patterns in images, which are critical for accurately distinguishing between subtle differences in medical imagery. Unlike CNNs, which use convolutional layers to extract these features, DNNs rely solely on fully connected layers, leading to poorer performance in tasks requiring high spatial resolution and detailed feature extraction.

CNN 1.0 models perform better than DNNs but still do not match the performance compared to deeper architectures like CNN 1.1 and transfer learning models. The unmasked, no DA, Gray, 64x64 configuration achieves a decent accuracy of 0.88, but the performance does not significantly improve with larger image sizes (299x299), which suggests that this architecture may not fully capitalize on the additional pixel information. The masked configurations performed notably worse, indicating a higher sensitivity to image masking.

Finally, CNN 1.1 models and transfer learning architectures demonstrate the best performance overall. The unmasked, DA, Gray, 256x256 configuration of CNN 1.1 achieved an accuracy of 0.94, comparable to the top-performing transfer learning models. Both approaches benefit significantly from data augmentation, which not only improves accuracy but also helps mitigate overfitting. Additionally, CNN 1.1's masked configurations still perform well, although not quite as robustly as the unmasked versions, indicating a well-balanced architecture capable of handling various preprocessing techniques effectively.

Table 4.4: Results overview of tested deep learning models

Preprocessing	Model type	accuracy score	F1-Score				Recall		execution time (min)
			Normal	COVID	Lung Opacity	Viral Pneumonia	macro average	COVID	
Masked, no DA, RGB, 224*224	ResNet50-SVM	0.87	0.90	0.80	0.84	0.91	0.86	0.73	10
Unmasked, no DA, RGB, 224*224	ResNet50-SVM	0.91	0.92	0.92	0.87	0.95	0.91	0.87	12
Masked, DA, RGB, 224*224	VGG16	0.90	0.92	0.85	0.87	0.96	0.90	0.80	60
Masked, DA, RGB, 224*224	ResNet50	0.90	0.92	0.82	0.87	0.96	0.89	0.75	47
Masked, DA, RGB, 240*240	EfficientNetB1	0.90	0.92	0.85	0.88	0.93	0.89	0.81	24
Unmasked, DA, RGB, 240*240	EfficientNetB1	0.94	0.95	0.96	0.92	0.97	0.95	0.93	34
Unmasked, no DA, Gray, 64*64	DNN 1.0	0.83	0.86	0.79	0.78	0.91	0.83	-	1
Unmasked, no DA, Gray, 299*299	DNN 1.0	0.81	0.85	0.78	0.74	0.90	0.82	-	1
Unmasked, no DA, Gray, 299*299	DNN 1.1	0.84	0.87	0.81	0.79	0.89	0.84	-	182
Unmasked, no DA, Gray, 64*64	CNN 1.0	0.88	0.84	0.90	0.89	0.95	0.90	-	1
Unmasked, no DA, Gray, 299*299	CNN 1.0	0.85	0.81	0.87	0.88	0.93	0.87	-	35
Masked, no DA, Gray, 64*64	CNN 1.0	0.78	0.85	0.54	0.76	0.89	0.76	-	1
Masked, no DA, Gray, 299*299	CNN 1.0	0.72	0.82	0.45	0.68	0.80	0.69	-	3
Masked, no DA, RGB, 256*256	CNN 1.1	0.89	0.92	0.85	0.86	0.93	0.89	0.88	25
Masked, DA, RGB, 256*256	CNN 1.1	0.91	0.93	0.88	0.89	0.95	0.91	0.85	56
Masked, DA, Gray, 256*256	CNN 1.1	0.92	0.93	0.89	0.89	0.95	0.92	0.88	60
Unmasked, DA, Gray, 64*64	CNN 1.1	0.91	0.93	0.92	0.89	0.94	0.92	0.87	9
Unmasked, DA, Gray, 256*256	CNN 1.1	0.94	0.94	0.96	0.91	0.95	0.94	0.94	31

4.2.2.2. Number of Channels

The role of grayscale images is particularly significant when comparing models that differ solely in the channel configuration, as it allows for a clearer understanding of the impact of input representation on performance. When comparing the performance of the Masked, DA, RGB, 256*256 and Masked, DA, Gray, 256*256 configurations of CNN 1.1, it is evident that

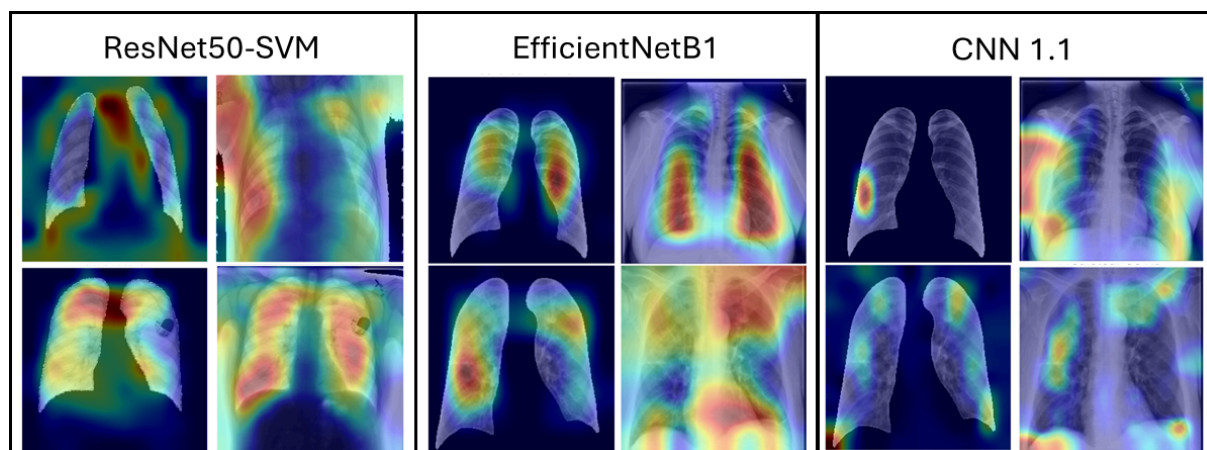
the grayscale model exhibits slightly better performance. The RGB model achieved an accuracy of 0.91 and a macro average F1-score of 0.95, while the grayscale model achieved an accuracy of 0.92 with a macro average F1-score of 0.92. Both models demonstrated strong recall scores for the COVID class, with the RGB model at 0.91 and the grayscale model at 0.92. Since both RGB and grayscale images originate from X-ray data, where all channels share the same pixel values, the use of RGB may complicate the analysis without providing substantial benefits. The marginal improvements seen with grayscale suggest that simpler models leveraging fewer channels can be equally effective or even superior. Additionally, it is important to note that hybrid and transfer learning models typically require specific input shapes and channel configurations, often favoring RGB input. Therefore, the differences between grayscale and RGB were not further investigated for these types of models, as the focus here was primarily on the CNN configurations.

4.2.2.3. Masking

The impact of masking is particularly noteworthy when comparing models that differ only in the masking component. For the Masked, no DA, RGB, 224*224 configuration, the ResNet50-SVM model achieved an accuracy of 0.87 and a macro average F1-score of 0.91. In contrast, the Unmasked, no DA, RGB, 224*224 configuration scored higher with an accuracy of 0.91 and a macro average F1-score of 0.95. This indicates that unmasked images perform slightly better overall. However, the masked model still demonstrates good performance and contributes significantly to feature extraction. In the context of grayscale images, comparing Masked, DA, Gray, 256*256 with Unmasked, DA, Gray, 256*256, the masked model achieved an accuracy of 0.92, while the unmasked version scored higher at 0.94. This further reinforces the observation that unmasked models generally perform better, while masked models still deliver competitive results, especially when enhanced by data augmentation techniques.

Furthermore, Grad-CAM visualizations (Figure 4.3) revealed notable differences in feature detection between the masked and unmasked versions. For both the ResNet50-SVM and CNN 1.1 models, masked and unmasked versions highlighted features outside the lung area, with extracted features appearing somewhat random. This suggests that in these models, masking does not have a substantial impact on guiding the model's focus to the lung region.

Figure 4.3: Grad-CAM images for masked (left) and unmasked (right) versions of ResNet50-SVM, EfficientNetB1 and CNN 1.1 models



However, for the transfer learning approach using EfficientNetB1, a clearer distinction was observed. The masked images led to feature detection predominantly within the lung area,

whereas unmasked images resulted in more diffuse and less lung-focused feature maps. This suggests that in the case of transfer learning, masking seems to enhance the model's ability to focus on relevant regions (i.e., the lungs), demonstrating that masking can influence feature extraction in specific architectures like EfficientNetB1.

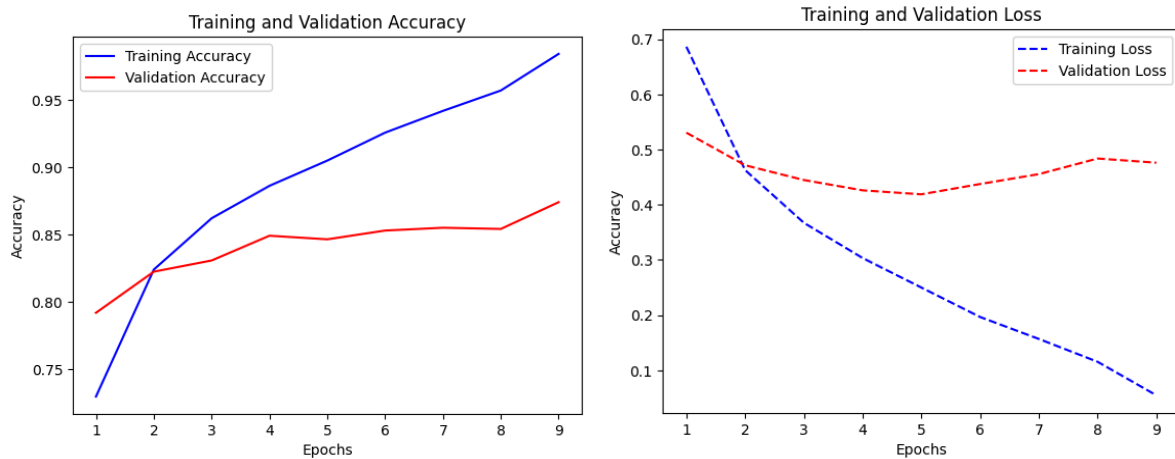
In conclusion, while unmasked models generally outperform their masked counterparts, masking in transfer learning appears to guide the model's attention more effectively to the lung area, making it a potentially useful technique in improving interpretability in specific contexts.

4.2.2.4. Data Augmentation

The use of Data Augmentation (DA) in the Masked, DA, RGB, 256*256 configuration using CNN 1.1 resulted in an accuracy of 0.91, with a macro average F1-score of 0.95. In comparison, the Masked, no DA, RGB, 256*256 configuration with the same model type achieved a slightly lower accuracy of 0.89 and a macro average F1-score of 0.93. When comparing the two versions of CNN 1.1 with and without data augmentation, it was observed that overfitting could be significantly reduced, demonstrating that data augmentation was more effective than other techniques aimed at addressing overfitting, such as increasing the dropout rate and introducing regularization. This improvement in performance highlights the effectiveness of data augmentation in enhancing model robustness.

It is also noteworthy that other models, including DNN 1.0, DNN 1.1, and CNN 1.0, exhibited strong signs of overfitting (Figure 4.4), underscoring the advantages of using data augmentation for improved classification performance in detecting COVID and other lung conditions.

Figure 4.4: Accuracy and loss by epoch of CNN 1.0



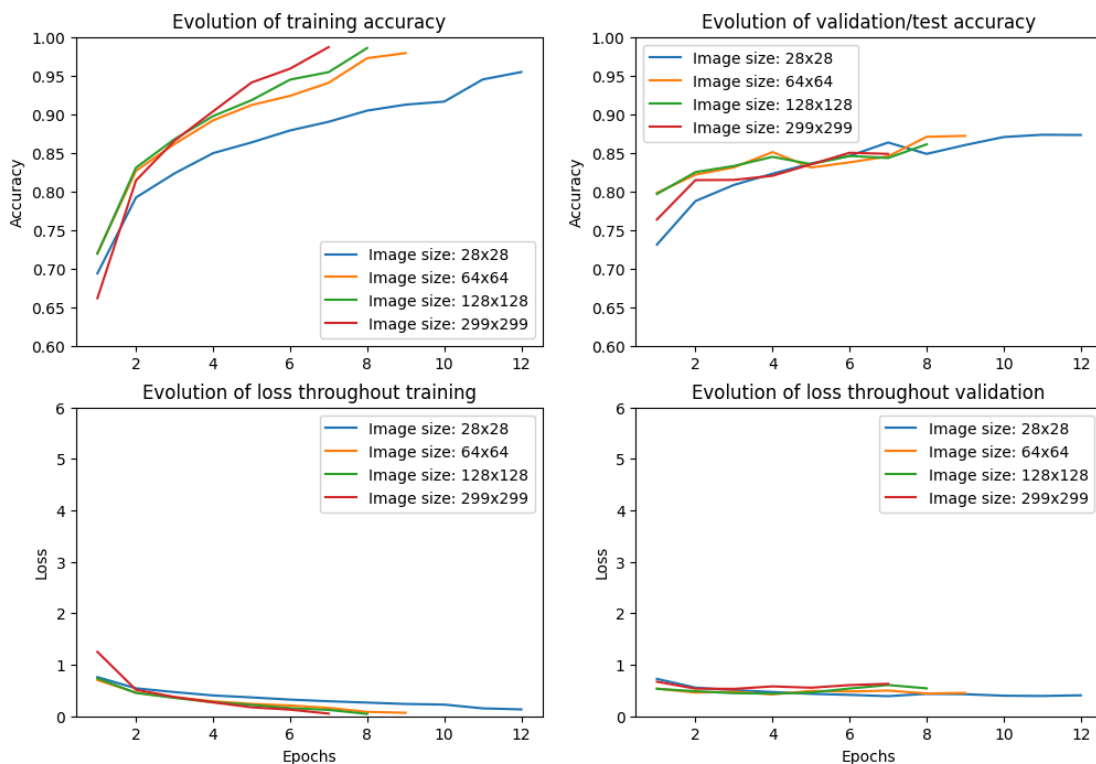
4.2.2.5. Image Size

The impact of image size on model performance is evident across various configurations, with larger image sizes generally expected to contribute to improved accuracy and F1-scores. However, this effect was not consistent across all models. For instance, in the DNN 1.0 models, the Unmasked, no DA, Gray, 64*64 configuration achieved an accuracy of 0.83, while the Unmasked, no DA, Gray, 299*299 configuration slightly underperformed with an accuracy of 0.81. This suggests that for DNN 1.0, increasing the image size from 64x64 to

299x299 did not lead to a performance gain, potentially due to overfitting or increased model complexity.

In the CNN 1.0 models, larger image sizes did not consistently lead to better performance either. The Unmasked, no DA, Gray, 64*64 configuration achieved an accuracy of 0.88, whereas increasing the image size to 299x299 resulted in a slightly lower accuracy of 0.85. This indicates that, for CNN 1.0, the larger image size may have introduced additional complexity without providing a sufficient benefit in capturing more detailed features, leading to a slight decrease in performance (Figure 4.5).

Figure 4.5: Accuracy and loss by epoch of CNN 1.0 for different image sizes



However, when data augmentation was applied in the CNN 1.1 models, the results were more favorable. The Unmasked, DA, Gray, 64*64 configuration achieved an accuracy of 0.91, and this further improved to 0.94 when the image size was increased to 256x256. This demonstrates that CNN 1.1, with its different architecture and the use of data augmentation, was able to leverage the larger image size more effectively, resulting in better overall performance.

On the other hand, for Masked, no DA configurations, larger image sizes did not improve performance. In the CNN 1.0 model with Masked, no DA, Gray, 64*64, the accuracy was 0.78, but increasing the image size to 299x299 actually decreased the accuracy to 0.72. This suggests that when masking is applied, the benefits of using larger images may be offset by the loss of contextual information in the masked regions, which the increased image resolution cannot fully compensate for.

Overall, while larger image sizes can be beneficial, particularly in models like CNN 1.1 with data augmentation, our results indicate that this is not universally the case, especially when other factors such as model architecture and masking are involved.

4.2.2.6. Activation Function and Batch Size

The impact of batch size and activation functions on model performance was also evaluated on CNN 1.0. Smaller batch sizes, such as 50, did not result in a significant performance difference compared to the larger batch sizes, indicating that batch size has minimal effect on overall accuracy and F1-Scores in this context. Regarding activation functions, tanh and sigmoid, performed slightly below the ReLU-based CNN 1.0 model. This suggests that ReLU remains the preferred activation function for this task, as it better supports model convergence and performance across all classes.

4.2.2.7. Summary

In our analysis of various model configurations for detecting COVID and other lung conditions, we prioritized Transfer Learning with EfficientNetB1 and CNN 1.1 due to their consistently superior performance metrics. Both models demonstrated very good results, effectively capturing the intricate features of medical images and yielding high classification accuracy. EfficientNetB1, leveraging unmasked images and data augmentation, showed very good accuracy, while CNN 1.1 also performed well under similar configurations. These results highlight the effectiveness of both approaches in handling complex medical imagery.

In addition, the analysis revealed that the CNN 1.1 models utilizing grayscale images achieved slightly better results than those using RGB images. This suggests that fewer channels may be less computationally intensive while maintaining comparable performance levels. As grayscale images can often simplify the analysis without compromising accuracy, this configuration appears to be a favorable choice for our models.

Our findings indicate that the use of masking in model configurations may be more detrimental than beneficial. The masked models generally underperformed compared to their unmasked counterparts, which may be attributed to the specific masking approach employed. Therefore, we have decided to favor unmasked images in our modeling strategy.

Data augmentation emerged as a critical factor in enhancing model robustness and reducing overfitting. The improved performance in configurations with data augmentation reinforces its necessity in training models effectively, especially given the complexities associated with medical imaging tasks.

While the effect of image size on model performance did not prove to be significantly influential across all configurations, we noted that higher image sizes, particularly in the case of CNN 1.1, yielded better results when full-sized images were used. This indicates that, for CNN 1.1, larger image dimensions can enhance the model's ability to leverage additional pixel information, thus improving classification accuracy.

In summary, our strategic decisions are driven by the clear advantages observed in using transfer learning and CNN 1.1 architectures, the effectiveness of unmasked images, the importance of data augmentation, and the beneficial effects of higher image sizes for CNN 1.1 models. These factors collectively contribute to a robust modeling approach for detecting COVID and other lung conditions.

4.2.3. Detailed Results of the Two Best Models

We will now present a detailed comparison of the two best-performing models: Transfer Learning with EfficientNetB1 and CNN 1.1. Both models have shown strong potential, and their architecture, training processes, and performance metrics will be discussed in depth.

4.2.3.1. Model 1 - Transfer Learning with EfficientNetB1

For the first model, we utilized the EfficientNetB1 architecture for transfer learning to classify chest X-ray images into four categories: Normal, COVID-19, Lung Opacity, and Viral Pneumonia. EfficientNet is a family of convolutional neural networks that achieve state-of-the-art performance while being more computationally efficient than previous architectures. This model employs transfer learning by leveraging pre-trained weights from ImageNet, allowing us to build upon the knowledge gained from a vast dataset.

The model was built using TensorFlow and Keras. Initially, the EfficientNetB1 model was loaded with pre-trained weights, excluding the top classification layer to adapt it for our specific task. All layers of the EfficientNetB1 base model were frozen to prevent their weights from being updated during the initial training phase. This approach allowed us to focus on training the newly added layers while leveraging the feature extraction capabilities of the pre-trained model.

The classification head of the model consists of several layers, starting with a Global Average Pooling layer that reduces the spatial dimensions of the feature maps from the EfficientNetB1 model into a single vector for each feature map, effectively summarizing the features. Following this, three fully connected Dense layers were added, each accompanied by Batch Normalization and Dropout layers to mitigate overfitting. These Dense layers contain 1024, 512, and 256 units, respectively, with a regularization term applied (L2 regularization) to further prevent overfitting. The final layer is a Dense layer with a softmax activation function, outputting probabilities for the four classes.

This architecture, combined with transfer learning techniques, allowed for effective classification of chest X-ray images, providing a strong foundation for further evaluation and refinement.

Training Process

The training process for the transfer learning model based on EfficientNetB1 involved several key steps to ensure effective learning from the dataset. Initially, the dataset was split into training and validation sets, with 80% allocated for training and 20% for validation. This stratified split ensured a balanced representation of the four classes: Normal, COVID-19, Lung Opacity, and Viral Pneumonia.

Data augmentation techniques, including flipping, rotation, zooming, and brightness adjustments, were applied to increase the diversity of the training dataset as described in 4.2.1.1. The EfficientNetB1 model was initialized without its top layers, focusing on transferring learned features from ImageNet rather than training the entire network from scratch. The last 31 layers of the base model were unfrozen from the start, allowing the model to adjust its pre-trained weights based on the new dataset.

Following the EfficientNetB1 base, fully connected layers were added, including Global Average Pooling, Dense layers with dropout for regularization, and a final softmax layer to output probabilities for each class. The model was compiled using the Adam optimizer and

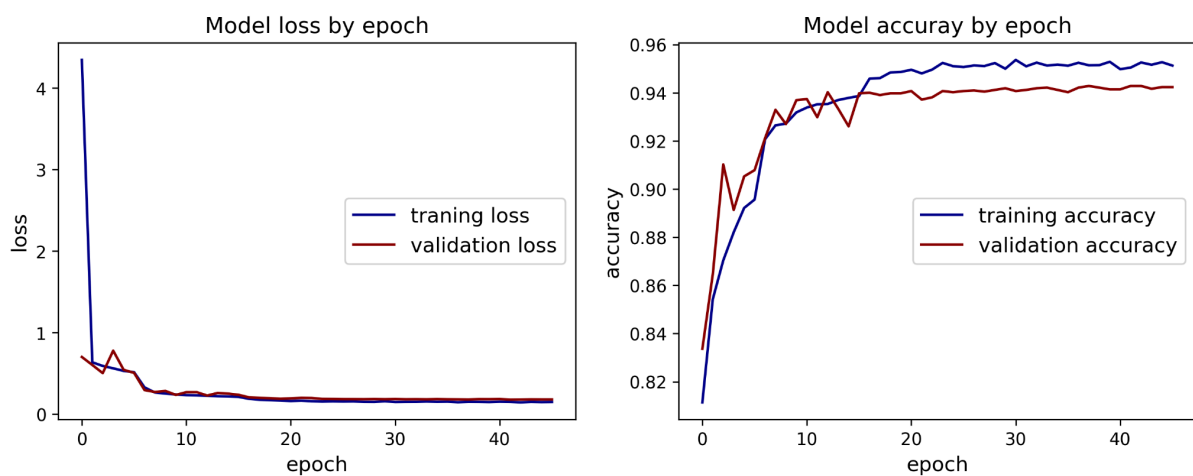
sparse categorical cross-entropy loss function, suitable for multi-class classification tasks. Training was conducted for up to 50 epochs, with early stopping based on validation performance.

The total modeling time was approximately 34 minutes, during which accuracy and loss metrics were used to track the model's performance.

Model Performance Metrics

The performance of the transfer learning model based on EfficientNetB1 was evaluated using several metrics, including training loss, validation loss, training accuracy, and validation accuracy over the course of 46 epochs (Figure 4.6).

Figure 4.6: EfficientNetB1 results - model loss and accuracy by epoch



Throughout the training process, the model exhibited a significant decrease in both training and validation loss, indicating improved learning and fitting to the dataset. The training accuracy consistently increased, reaching a peak of approximately 95.2%, while the validation accuracy showed a similar upward trend, stabilizing around 94.2%.

The training loss dropped from an initial high of 4.35 to around 0.15 by the final epoch, demonstrating effective convergence. The validation loss, initially at 0.70, also decreased to 0.18, further validating the model's generalization capabilities.

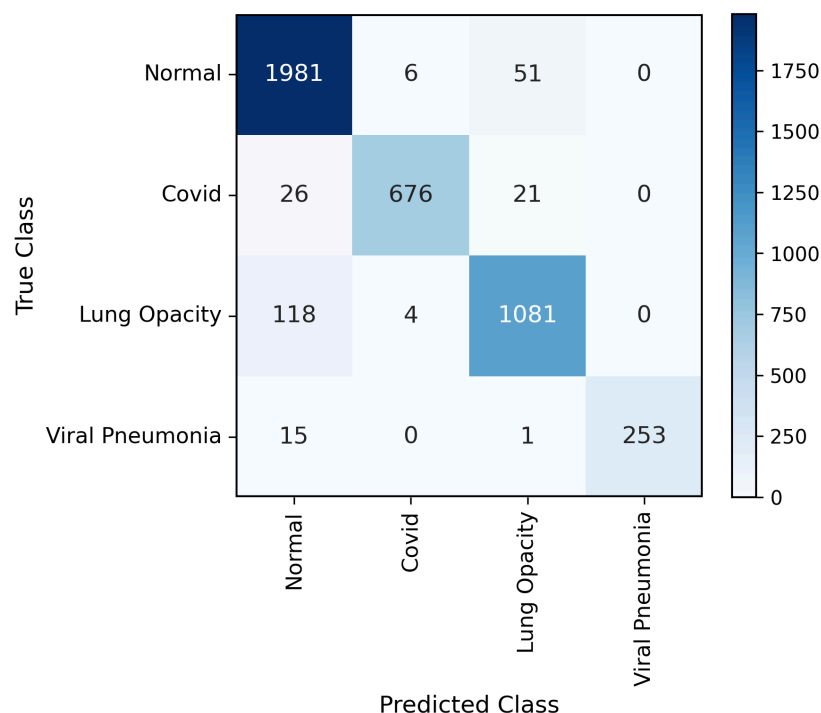
These results highlight the model's effectiveness in classifying chest X-ray images, demonstrating its potential for real-world applications in medical image analysis.

Evaluation on Validation Data

The evaluation of the transfer learning model on the validation dataset yielded very good results, with an overall accuracy of approximately 94.28% (Table 4.4). This indicates the model's strong capability to accurately classify chest X-ray images into the defined categories. Focusing on the COVID-19 class and Viral Pneumonia class, the confusion matrix (Figure 4.7) reveals that the model correctly identified 676 instances of COVID-19, with only 26 false negatives classified as Normal and 21 as Lung Opacity. This demonstrates a high precision for the COVID-19 category, reflecting the model's effectiveness in detecting this critical condition.

For Viral Pneumonia, the model achieved perfect precision, correctly identifying all 253 instances, while having only 15 false negatives classified as Normal and 1 as Lung Opacity. The classification report further highlights the model's strong performance for these classes, with COVID-19 yielding a precision of 0.99 and a recall of 0.93, resulting in an F1-score of 0.96. Similarly, for Viral Pneumonia, the precision is 1.00, with a recall of 0.94 and an F1-score of 0.97. The model's ability to minimize false positives in both COVID-19 and Viral Pneumonia classifications emphasizes its reliability as a diagnostic tool. Overall, the high accuracy, precision, and recall values indicate that the model is well-suited for identifying these critical lung conditions in chest X-ray images, making it potentially a valuable asset for medical image analysis.

Figure 4.7: EfficientNetB1 results - confusion matrix



In addition to these quantitative metrics, we utilized Grad-CAM (Gradient-weighted Class Activation Mapping) (Figure 4.8) to visualize the areas of the images that the model focused on during classification. Interestingly, the Grad-CAM images revealed that the model did not always highlight regions within the lung fields; instead, it sometimes focused on areas outside the lungs. This observation raises questions about the model's decision-making process and indicates that it may have learned to identify features related to disease that are not confined to the traditional boundaries of lung anatomy.

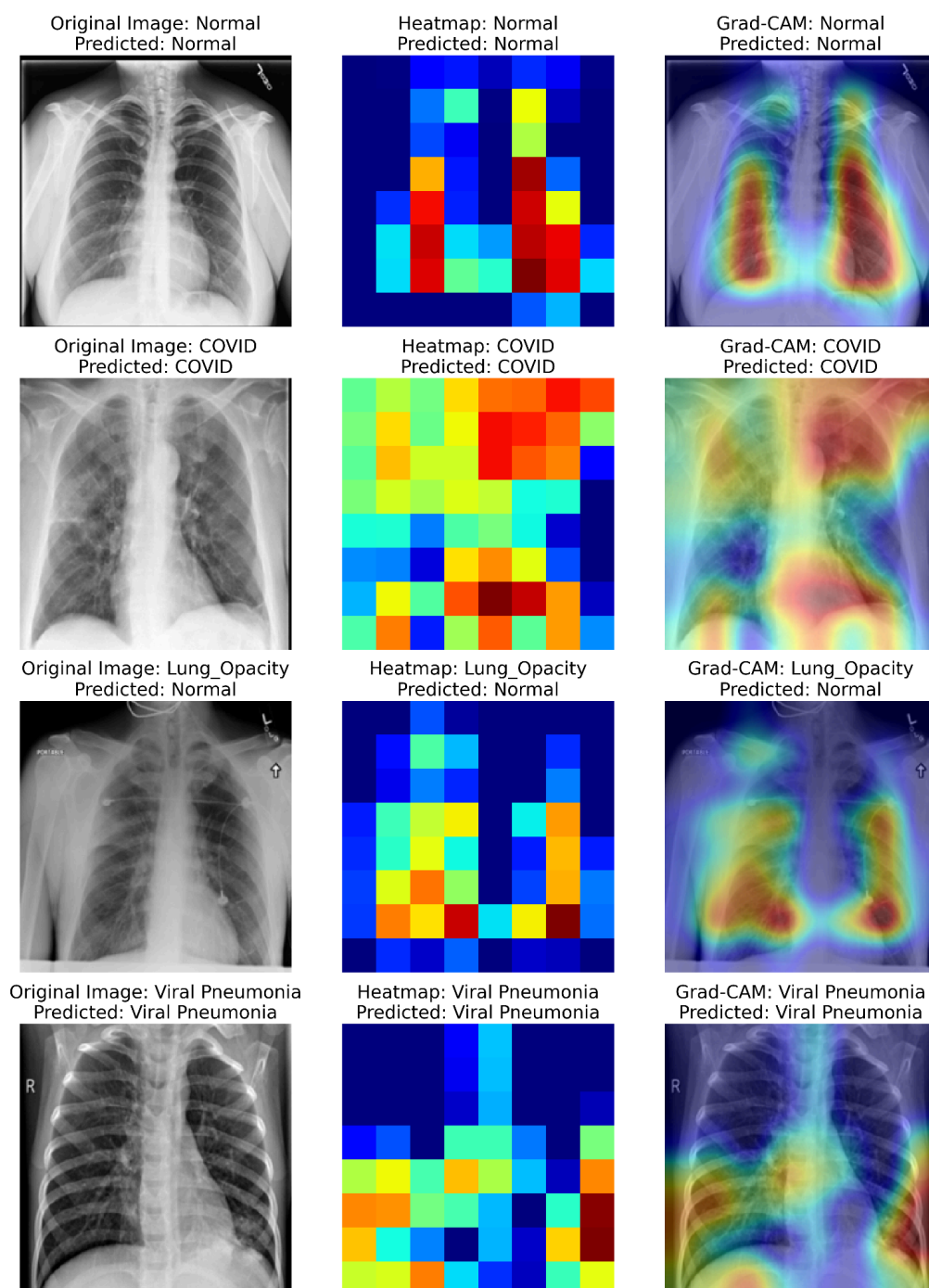
This observation can be attributed to several potential explanations. First, it is possible that the model is picking up on contextual cues or artifacts present in the images, such as surrounding anatomical structures or external features that may correlate with the presence of COVID-19 or Viral Pneumonia. These cues might not directly relate to the pathological features within the lungs but can still provide valuable information for classification.

Second, the presence of noise or inconsistencies in the training data may influence the model's attention. If certain features or regions outside the lungs are consistently associated with specific conditions in the dataset, the model may learn to focus on these areas rather than strictly lung-related features.

Lastly, it may be indicative of the complex nature of these diseases, where symptoms or signs can manifest in a way that is not exclusively confined to the lungs. For instance, pneumonia can lead to systemic changes that are reflected in other areas of the body, and the model may be identifying these broader patterns.

Overall, the Grad-CAM results suggest that while the model performs well in classifying COVID-19 and Viral Pneumonia, it may rely on factors beyond traditional lung imaging cues, highlighting the importance of understanding model interpretability in clinical applications. This finding also emphasizes the need for further investigation to enhance the model's focus on relevant regions and improve its clinical utility.

Figure 4.8: EfficientNetB1 - Heatmap and Grad-CAM images per case on validation dataset



4.2.3.2. Model 2 - CNN 1.1

The second model is a Convolutional Neural Network (CNN) designed to classify medical images into four categories: Normal, COVID, Lung Opacity, and Viral Pneumonia. The input images are grayscale and resized to 256x256 pixels with one channel. In summary, the 256x256 size was selected to balance model performance and training efficiency, ensuring the model can learn effectively from the available data without overwhelming computational resources.

The model begins with an input layer that accepts images of size (256, 256, 1). It then applies a series of convolutional blocks for feature extraction. The first block includes a 2D convolutional layer with 32 filters of size (3x3), followed by batch normalization and max-pooling to reduce the spatial dimensions. The second block increases the filter count to 64, again applying ReLU activation, batch normalization, and max-pooling. In the third block, the filter count is further increased to 128, while the fourth and fifth blocks use 256 and 512 filters, respectively. Each of these blocks follows the same structure: convolution, batch normalization, and max-pooling.

Once the convolutional layers have extracted features, the output is flattened and passed through two fully connected layers. The first dense layer contains 512 units and is followed by a 50% dropout to prevent overfitting, as well as L2 regularization with a penalty coefficient of 0.001. The second dense layer contains 256 units and also includes dropout and L2 regularization. Finally, the model has an output layer with 4 neurons corresponding to the four target classes, using a softmax activation function to output a probability distribution for classification.

Training Process

The model was trained using the following pipeline. Before training, the dataset was split into two subsets: 80% of the data was used for training, and 20% was used for validation. The images were grayscale, resized to 256x256 pixels, and passed through data augmentation techniques (see 4.2.1.1) to increase variability and prevent overfitting.

The model was compiled using the Adam optimizer with a learning rate of 0.001, chosen for its adaptive learning rate capabilities and robustness in handling sparse gradients. The loss function used was sparse categorical cross-entropy, which is suitable for multi-class classification tasks, and accuracy was the primary performance metric.

Training was performed in batches of 64 images to ensure efficient GPU utilization while keeping the memory footprint manageable. Additionally, the TensorFlow `tf.data.AUTOTUNE` functionality was used to optimize data loading, prefetching batches during training to maximize computational efficiency.

Two callbacks were implemented to monitor the training process and ensure optimal model performance. The `ReduceLROnPlateau` callback automatically reduced the learning rate by a factor of 0.1 if the validation loss plateaued for three consecutive epochs, allowing the model to fine-tune weights in later stages of training. Furthermore, an `EarlyStopping` callback was employed to halt training if the validation loss did not improve for four consecutive epochs, preventing overfitting by restoring the best model weights when the model stopped improving.

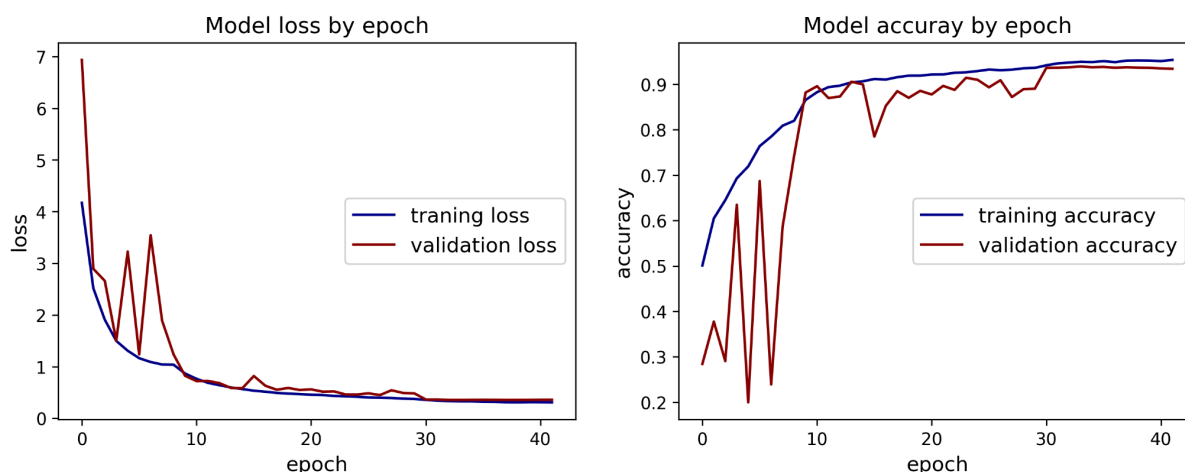
The model was trained for up to 50 epochs, although early stopping was triggered after validation loss plateaued, shortening the actual number of epochs to 42. The overall training

time was 32 minutes. Each epoch's training loss and accuracy were monitored and compared to the validation loss and accuracy to track overfitting or underfitting (Figure 4.7). Additionally, the ReduceLROnPlateau callback helped fine-tune the model when the learning rate needed adjustment, ensuring that the training was adaptive as the model progressed through different stages of learning.

Model Performance Metrics

The model's performance was evaluated over 42 epochs (Figure 4.9). During the initial epochs, the model experienced a notable disparity between training and validation performance. At epoch 1, the training loss was 4.17, and the validation loss was significantly higher at 6.93, with training accuracy at 50.1% and validation accuracy at 28.4%. This gap indicates that the model was still in the early stages of learning, struggling to generalize to the validation set.

Figure 4.9: CNN 1.1 results - model loss and accuracy by epoch



As the epochs progressed, the training accuracy consistently improved, reaching 94.1% by epoch 31, with a corresponding validation accuracy of 93.6%. The validation loss also decreased, stabilizing at around 0.35 by epoch 39. This suggests that the model successfully learned to distinguish between the classes, particularly after the early learning phases when it had a more substantial validation loss fluctuation.

The highest validation accuracy, 94.9%, was achieved around epoch 33, which reflects the model's ability to generalize well to unseen data. After this point, both the training and validation losses and accuracies converged, indicating a stable training process. However, some signs of potential overfitting emerged as the training accuracy continued to rise slightly, while validation accuracy plateaued or marginally decreased in the final epochs.

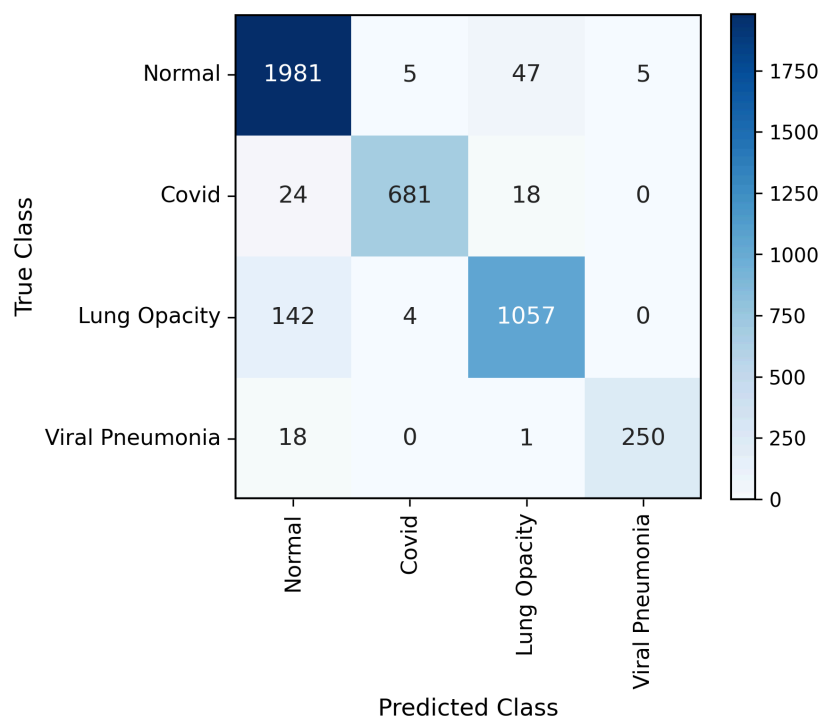
Overall, the model demonstrates strong learning capabilities, with high performance across both the training and validation sets. The final validation accuracy of 93.4% and relatively low validation loss point to a well-trained model capable of effective classification.

Evaluation on Validation Data

The model's performance was evaluated on a separate validation dataset, achieving an overall accuracy of approximately 93.76% (Table 4.4).

Focusing on the classes of particular interest, COVID-19 and Viral Pneumonia, the confusion matrix (Figure 4.10) revealed that the model correctly classified 681 instances of COVID-19, with 24 misclassified as normal and 18 as lung opacity. For Viral Pneumonia, the model accurately identified 250 instances, with only 18 misclassified as normal and 1 as lung opacity. These results indicate that the model performs effectively in distinguishing between these two critical classes.

Figure 4.10: CNN 1.1 results - confusion matrix



The classification report further underscores the model's capabilities for these classes. For COVID-19, the precision was 0.99, indicating a high level of correctness in identifying positive cases, while the recall was 0.94, showing that the model successfully detected 94% of actual COVID-19 cases. This resulted in an F1-score of 0.96, reflecting an excellent balance between precision and recall.

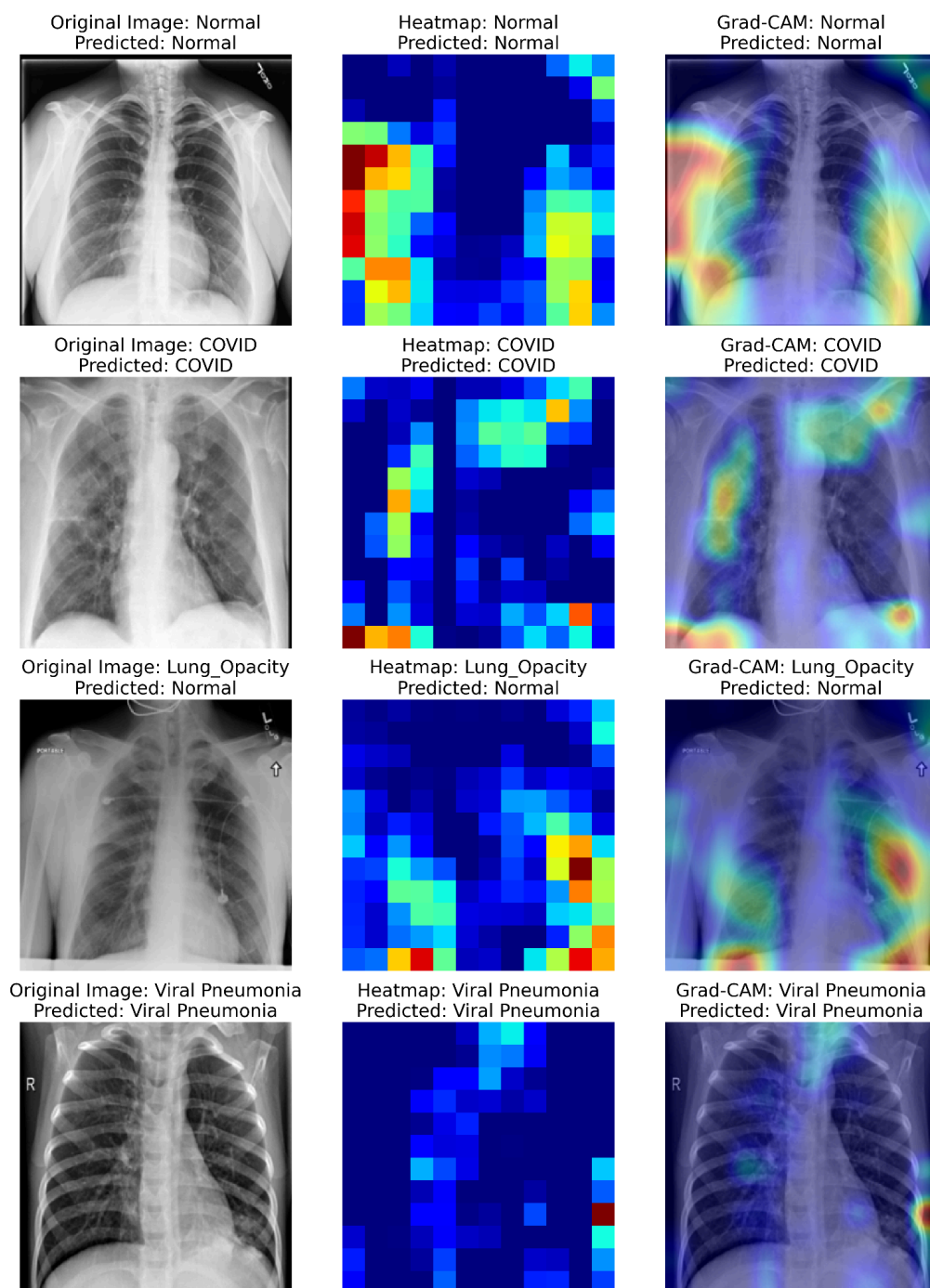
In the case of Viral Pneumonia, the model achieved a precision of 0.98, meaning it correctly identified nearly all positive instances, and a recall of 0.93, indicating that it captured 93% of actual cases. The F1-score of 0.95 demonstrates that the model performs reliably for this class as well.

Overall, the evaluation highlights the model's effectiveness in classifying COVID-19 and Viral Pneumonia, with high precision, recall, and F1-scores.

To gain insights into the decision-making process of the model, we generated Grad-CAM images (Figure 4.11) that visualize the areas of the input images considered most important during predictions. The resulting heatmaps indicated that the regions deemed significant are not always confined to the lung areas, which aligns with previous observations regarding the transfer learning model. For possible explanations see section 4.2.3.1 (Evaluation on Validation Data). This finding underscores the need for a deeper understanding of model interpretability, particularly in clinical contexts. While the model shows robust classification performance for COVID-19 and Viral Pneumonia, the Grad-CAM results suggest it may be

relying on additional factors beyond typical lung imaging cues. Understanding these influences is essential for enhancing the model's clinical relevance.

Figure 4.11: CNN 1.1 - Heatmap and Grad-CAM images per case on validation dataset



4.2.3.3. Comparison of the two Models

Both models achieved high accuracy, with Model 1 (Transfer Learning with EfficientNetB1) slightly outperforming Model 2 (CNN 1.1) (94.28% vs. 93.76%). Model 1 demonstrated particularly strong precision and recall in classifying Normal and COVID-19 cases, while both models performed similarly for Lung Opacity and Viral Pneumonia. The confusion matrices reveal that Model 1 had fewer false positives and false negatives across categories,

indicating a more robust classification performance overall. In terms of macro averages, both models performed comparably, with only minor differences in recall and F1-scores.

Additionally, the modeling times were similar, with Model 1 requiring 34 minutes and Model 2 taking 32 minutes. Moreover, Grad-CAM images for both models indicated that the detected features are not always confined to the lung areas. This observation highlights the models' reliance on factors beyond traditional lung imaging cues, emphasizing the need for further investigation into their interpretability and clinical utility.

5. Conclusion

For decision-making regarding potential deployment, we considered the following aspects: accuracy, computational efficiency, interpretability, and clinical utility.

In terms of performance, Model 1 (EfficientNetB1) slightly outperforms Model 2 (CNN 1.1) with an accuracy of 94.28% compared to 93.76%. Model 1 also shows stronger precision and recall, particularly for Normal and COVID-19 cases. Although both models perform similarly for "Lung Opacity" and "Viral Pneumonia," Model 1 has fewer false positives and false negatives, making it the more reliable choice for accurate classification.

Both models have similar training times, with Model 1 requiring 34 minutes and Model 2 taking 32 minutes. However, Model 1 leverages transfer learning, utilizing pre-trained weights from EfficientNetB1, which makes it more computationally efficient and easier to adapt to new tasks. This gives Model 1 a slight edge in deployment scenarios, especially when computational resources may be limited.

When it comes to interpretability, both models' Grad-CAM visualizations show that they detect features outside of the lung areas, which suggests that they rely on external or contextual cues unrelated to the lungs. This introduces some uncertainty regarding the models' decision-making processes, which raises concerns about deploying either model in clinical settings without further investigation into their interpretability.

Though Model 2 is competitive in performance, there is more potential for improvement, particularly in its feature extraction process. On the other hand, Model 1 benefits from a robust feature extraction pipeline, thanks to the EfficientNetB1 architecture. While Model 2 may allow for more flexibility in refinement, Model 1 is already a more optimized solution, ready for deployment with fewer modifications needed.

When determining the best model for deployment, we considered key factors such as accuracy, computational efficiency, interpretability, and clinical utility. Both models show strong potential in identifying COVID-19 and Viral Pneumonia, which is vital for timely diagnosis and treatment. However, Model 1 (EfficientNetB1) offers superior accuracy, precision, and reliability, making it the more dependable choice, especially in clinical settings where accuracy is critical. While both models could benefit from refinement, Model 1's efficient feature extraction and robust performance make it the better candidate for immediate deployment.

6. Outlook

The outlook for the potential deployment of the algorithm reveals several critical areas for improvement that could significantly enhance its business case and overall effectiveness in clinical settings. As the algorithm stands, there are limitations within the data that, if

addressed, could improve the accuracy of disease detection and expand the utility of the model.

Data limitations and improvement of labeling

One of the primary limitations identified in the current model is its difficulty in accurately detecting cases of lung opacity. That is, lung opacity is frequently misclassified as a normal condition. Lung opacity refers to a degree of haziness on an X-ray that may indicate potential diseases beyond COVID-19 or Viral Pneumonia. However, the graphical descriptive statistics have shown that the pixel values between lung opacity and normal conditions are similar, leading to challenges in differentiating between these two classes.

One potential way to address this issue is to examine the quality of the primary data collection processes. There is a possibility that healthy lungs have been incorrectly classified as opacity cases or vice versa, leading to inaccuracies in the model's training data. Additionally, inconsistencies across different sources in defining what degree of haziness constitutes lung opacity could further contribute to the model's challenges in accurate classification. One option would be a cross-validation process with independent expert pulmonologists. An alternative would be to use self-supervised learning to detect inconsistencies in the labels among sources. Standardizing the label definitions and improving the labeling accuracy during data collection would help enhance the model's performance.

Future work could also explore integrating multimodal data such as patient history, symptom records, and laboratory results to improve diagnostic accuracy.

Model improvements

In addition to addressing the data limitations, several potential improvements to the model itself could enhance its performance. These improvements focus on fine-tuning the existing models and exploring alternative approaches to optimize results.

One general strategy to consider is experimenting with removing lung opacity as a class altogether. Given the challenges associated with accurately classifying this condition and its overlap with other classes, eliminating it might simplify the model's task and improve its overall accuracy. This approach could be particularly effective if lung opacity has been misclassified in the primary data generation, leading to confusion and reducing the model's performance.

For Transfer Learning models, one avenue for improvement involves experimenting with different base models. The current base model may not be the best fit for the specific characteristics of lung X-ray data. So far, we have tested EfficientNetB1, VGG16, and ResNet, but there may be other pre-trained models more closely aligned with the classification of lung X-ray images. These models could potentially learn more relevant features, leading to better performance. Additionally, we could explore unfreezing more layers or different layers beyond what has been shown above. By doing so, the model could learn more specific features from the lung X-ray images, potentially improving its ability to distinguish between different conditions. Another approach is to experiment with ensemble methods, such as model averaging or stacking different transfer learning models or an ensemble of models trained with different configurations. Averaging their predictions could result in more robust and accurate outcomes.

For the CNN model, another potential improvement lies in modifying the kernel size, an aspect that has not yet been explored. By experimenting with different kernel sizes, the model might better capture the fine details necessary for distinguishing between similar-looking conditions, such as lung opacity and normal lung images. Additionally, experimenting with adding more convolutional and pooling layers could be beneficial. A deeper network might capture more complex features, although it is essential to balance depth with the risk of overfitting.

Extending the labels to include further diseases

Currently, the model is capable of detecting only two specific lung diseases: Viral Pneumonia and COVID-19. All other lung-related conditions, such as lung cancer, heart failure, tuberculosis, sarcoidosis, and fibrosis among others, are grouped under the broad category of Lung Opacity. However, inter-class differences among these diseases could negatively impact the model's performance, as each condition may exhibit distinct characteristics in the lung image that cannot be adequately captured by a single class.

To improve the use case of the model in the longer run, it would be beneficial to extend its capabilities to detect a wider range of lung diseases. In clinical practice, a tool that can identify more than just two diseases would be far more valuable to healthcare providers. However, this would require further data collection efforts, focusing on obtaining a diverse set of images representing various lung conditions. However, such a data collection would be time-consuming and expensive. Thus, an alternative approach would be to use a smaller set of lung x-ray images with a broader range of diseases and apply self-supervised or semi-supervised learning on the existing data set. By expanding the model's training data to include these additional diseases, the algorithm could be refined to provide more accurate and comprehensive diagnostic support.

In summary, while the current model shows promise, addressing the data quality issues, fine-tuning the model and expanding its disease detection capabilities are crucial steps for its future deployment. By implementing these improvements, the model's potential to provide accurate and comprehensive diagnostic support in real-world applications could be greatly enhanced, ultimately contributing to better healthcare outcomes.